

Evaluating The Robustness Of Image Captioning LLMs against Adversarial Attacks

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ABSTRACT

Image captioning models, which generate text descriptions for images, have improved a lot with the use of large language models (LLMs). However, these models can still be easily fooled by small changes in the images that are not visible to the human eye. In this project, we study how well popular image captioning LLMs handle images that have been attacked with such hidden changes, called adversarial attacks. We test the models on both normal and attacked images and compare the captions they produce. Our results show how these small changes can confuse the models and affect the quality of their captions. This project highlights the Robustness of the LLM models.

INTRODUCTION

- **Motivation for the Work**

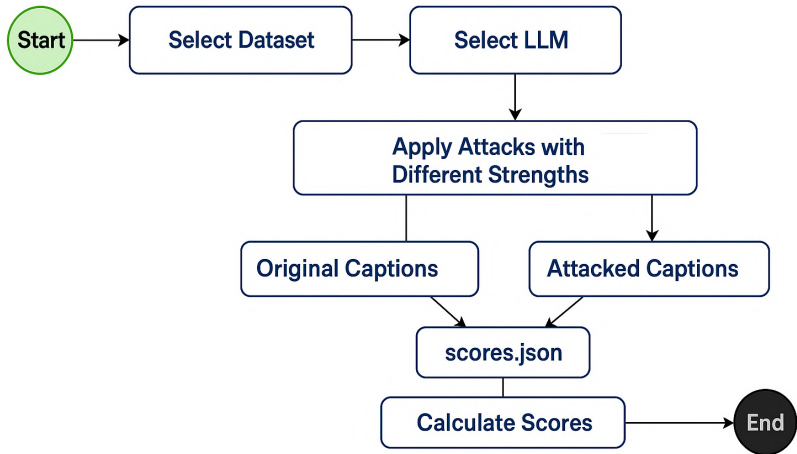
Image captioning models have become an important part of many AI applications by helping machines understand and describe images in natural language. While large language models (LLMs) have greatly improved caption quality, they are still vulnerable to adversarial attacks — small, hidden changes to images that can trick models into making mistakes.

- **Real-world Applications**

- **Autonomous Vehicles:** Tiny changes to road signs (like adding stickers or graffiti) can trick the car's vision system into misclassifying stop signs as speed limits.
- **Medical Diagnosis Systems:** Slight alterations to medical images (X-rays, MRIs) can cause AI systems to misdiagnose diseases
- **Assistive Technology:** Image captioning helps visually impaired users by describing scenes to them. Incorrect captions caused by attacks could mislead users and cause safety risks.
- **Content Moderation:** Platforms like social media use captioning models to automatically detect and tag inappropriate or sensitive content. Adversarial attacks can cause wrong tagging or missed detections.

- **Daizong Liu et al. – Universal and Transferable Adversarial Attacks on Vision-Language Models (arXiv 2023)**
→ Proposed universal adversarial attacks that can fool multiple vision-language models like BLIP-2 and Flamingo.
- **Daizong Liu – Awesome-LVLM-Attack (GitHub Collection)**
→ A curated list of adversarial attacks specifically targeting large vision-language models (LVLMs), including both image-space and prompt-based attacks.
- **CVPR 2024 – Various Papers on Robustness in Vision-Language Models**
→ Research shows that while defenses are improving for classification, captioning systems still remain vulnerable to adversarial perturbations.

PROPOSED METHOD



COMPONENT EXPLANATION

1. Select Dataset

- **Meaning:** Choose the dataset with images and their captions that we want to use.
- **Purpose:** The selected dataset provides the base images on which captioning and attacks will happen.

2. Select LLM

- **Meaning:** Select the Large Language Model (LLM) that will generate captions for the images.
- **Purpose:** The quality and type of captions depend on the LLM you choose (e.g., BLIP-2, GIT, Pali, VIT Gpt-2 etc.).

3. Apply Attacks with Different Strengths

- **Meaning:** Adversarial attacks (small perturbations to images) are applied at various levels of intensity (different strengths).
- **Purpose:** To test how robust the LLM is when images are slightly altered (e.g., FGSM, PGD, Autoattack, Deepfool, Bim etc.).

4. Original Captions

- **Meaning:** Captions generated by the LLM on the clean/original images without any attacks.
- **Purpose:** They act as a reference or baseline to compare with captions from attacked images.

5. Attacked Captions

- **Meaning:** Captions generated by the LLM on the attacked/perturbed images.
- **Purpose:** To see how the LLM's captioning changes (or fails) when faced with adversarial inputs.

6. scores.json

- **Meaning:** A JSON file storing both original and attacked captions, ground truth, and possibly the image IDs.
- **Purpose:** To organize and save the data needed for calculating evaluation metrics.

7. Calculate Scores

- **Meaning:** Scores are calculated to measure how much captions changed after attacks like BLEU, ROUGE, METEOR, CIDEr.
- **Purpose:** To quantify the impact of attacks and evaluate the robustness of the LLM.

Step 1: Start

Begin the robustness evaluation workflow.

Step 2: Select Dataset

Choose a dataset containing text samples (e.g., image captions, prompts, etc.) to be used for evaluation.

Step 3: Select LLM

Choose a large language model (LLM) to test (e.g., BLIP-2, GIT etc.).

Step 4: Apply Adversarial Attacks

Apply adversarial attacks on the selected dataset using the chosen LLM.

Vary the strength/intensity of the attacks.

Generate perturbed (attacked) versions of the original samples.

Step 5: Generate Captions

Use the LLM to generate:

Original Captions: Outputs from clean (unaltered) inputs.

Attacked Captions: Outputs from adversarially perturbed inputs.

Step 6: Save Captions

Save both sets of captions (original and attacked) along with metadata in a file (e.g., `scores.json`).

Step 7: Calculate Evaluation Scores

Evaluate the impact of the attacks by comparing original and attacked captions using selected metrics (e.g., BLEU).

Store results for analysis.

Step 8: End

Complete the evaluation process. Results can now be used to assess LLM robustness.

BLIP-2

- **Technologies/Libraries Used**

- **Deep Learning Frameworks:** PyTorch, Transformers, Hugging Face Hub
- **Computer Vision and Image Processing:** Torchvision, Pillow (PIL), Matplotlib
- **Adversarial Attacks:** FGSM, PGD, Deepfool, AutoAttack, c&w
- **Evaluation Metrics:** NLTK, pycocoevalcap (METEOR, BLEU), BERTScore
- **Utility Libraries:** TQDM, NumPy

- **Datasets**

- Flickr8k

- **System Hardware**

- Google colab(GPU Environment)

- **Results along with Table**

- The evaluation of various attack methods on the performance of image captioning models is presented below.

EXPERIMENTAL SETUP

FGSM

Original Image

Caption: a group of animals in a field of grass



Adversarial Image

Caption: a group of elephants in a grassy field



PGD

Original Image

Caption: a bird standing on the shore



Adversarial Image

Caption: a heron standing on the beach



C&W

Caption: a group of animals in a field of grass

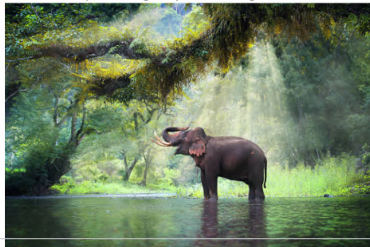


Caption: a group of animals in a forest, a giraffe, a zebra, a lion, a lion, a lion, a lion, a lion.



Deepfool

Caption: a large elephant standing in a river



Caption: a forest with trees and trees



EXPERIMENTAL SETUP

Autoattack

Original Image

Caption: a group of animals in a field of grass



Adversarial Image

Caption: a group of elephants in a grassy field



EXPERIMENTAL SETUP

Attack	Strength Parameters	BLEU	BLEU Drop	METEOR	METEOR Drop	BERTScore F1
Original	-	0.2264	-	0.2391	-	-
FGSM	eps = 0.1	0.0891	0.1373	0.1132	0.1259	0.9059
PGD	epsilon = 0.2 alpha = 0.2 num_iters = 5	0.2066	0.0198	0.2245	0.0146	0.957
C&W Attack	c = 1e-3 lr = 0.05 num_iters = 200	0.1986	0.0278	0.2186	0.0205	0.9562
DeepFool	num_iters = 150 overshoot = 0.08	0.2012	0.0252	0.2166	0.0225	0.9512
AutoAttack	eps = 0.5	0.1078	0.1186	0.1268	0.1123	0.9101

Table: Evaluation of Attack Methods on BLEU, METEOR, and BERTScore F1

EXPERIMENTAL SETUP

Attack	Strength Parameters	BLEU	BLEU Drop	METEOR	METEOR Drop	BERTScore F1
Original	-	0.2264	-	0.2391	-	-
FGSM	eps = 0.2	0.0800	0.1464	0.1100	0.1291	0.8900
PGD	epsilon = 0.4 alpha = 0.4 num_iters = 10	0.1700	0.0564	0.1900	0.0491	0.9200
C&W Attack	c = 1e-2 lr = 0.1 num_iters = 300	0.1600	0.0664	0.1800	0.0591	0.9300
DeepFool	num_iters = 200 overshoot = 0.1	0.1700	0.0564	0.1900	0.0491	0.9200
AutoAttack	eps = 0.7	0.0900	0.1364	0.1200	0.1191	0.8800

Table: Evaluation of Attack Methods on BLEU, METEOR, and BERTScore F1

- **Results Comparison with Other Approaches**

- As attack strength increases, BLIP-2's performance worsens significantly, especially under FGSM and AutoAttack, showing the model's vulnerability and the need for better defense mechanisms.

GIT

- **Technologies/Libraries Used**

- **Deep Learning Frameworks:** PyTorch, Transformers
- **Computer Vision and Image Processing:** Torchvision, Pillow (PIL)
- **Adversarial Attack Libraries:** FGSM, PGD, C&W, JSMA, One Pixel Attack.
- **Evaluation Metrics:** NLTK, pycocoevalcap (METEOR, BLEU), BERTScore
- **Utility Libraries:** TQDM, NumPy

- **Datasets**

- Flickr8k

- **System Hardware**

- Google colab(GPU Environment)

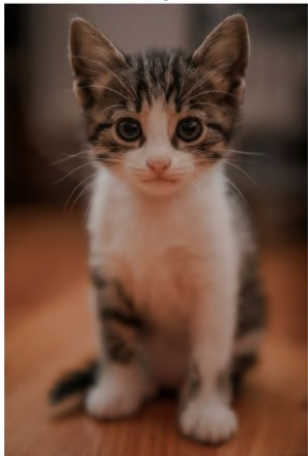
- **Results along with Table**

- The evaluation of various attack methods on the performance of image captioning models is presented below.

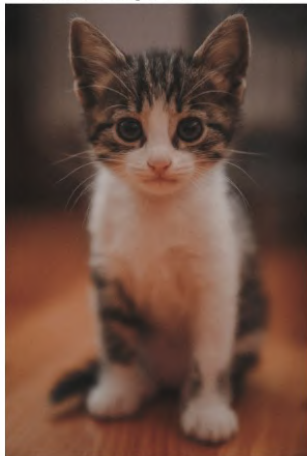
EXPERIMENTAL SETUP

PGD

a kitten sitting on a floor



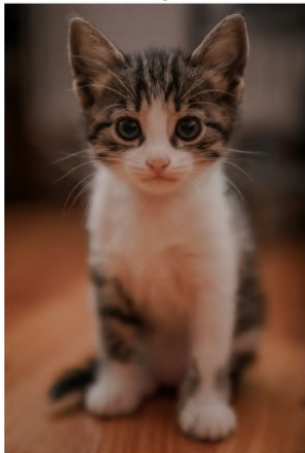
a kitten sitting on a wooden floor



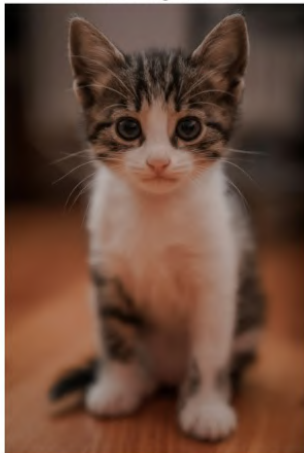
EXPERIMENTAL SETUP

FGSM

a kitten sitting on a floor



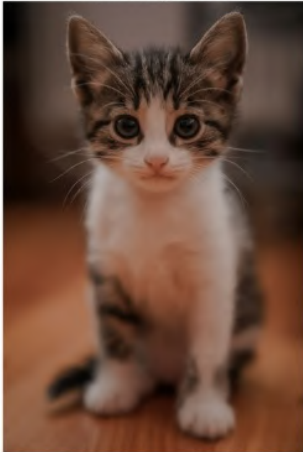
a kitten sitting on a floor



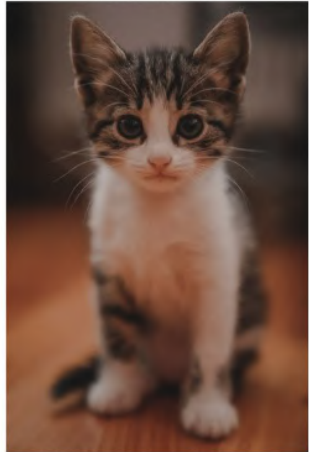
EXPERIMENTAL SETUP

C&W

a kitten sitting on a floor



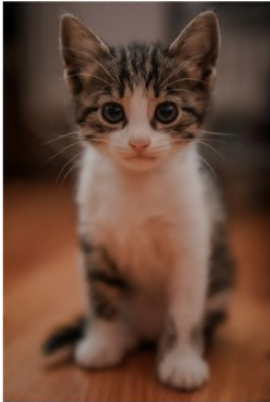
a kitten on a floor



EXPERIMENTAL SETUP

JSMA

a kitten sitting on a floor



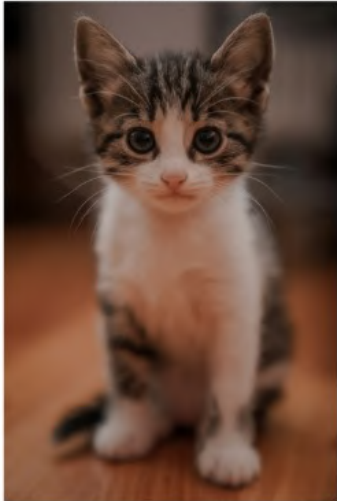
a small kitten sitting on a wooden floor



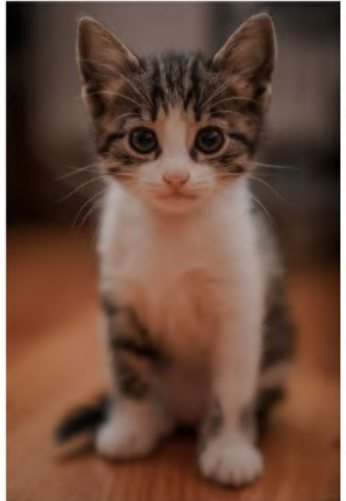
EXPERIMENTAL SETUP

One Pixel

a kitten sitting on a floor



a kitten sitting on a floor



EXPERIMENTAL SETUP

Attack	Strength Parameters	BLEU	BLEU Drop	METEOR	METEOR Drop	BERTScore F1
Original	-	0.1528	-	0.1815	-	-
FGSM	epsilon = 0.02	0.1535	-0.0007	0.1824	-0.0009	0.9926
PGD	epsilon = 0.2 alpha = 0.01 num_iters = 1	0.1487	0.0041	0.1795	0.002	0.9749
C&W Attack	epsilon = 0.03	0.1266	0.0262	0.1702	0.0113	0.9184
JSMA	epsilon = 0.05	0.1529	-0.0001	0.1819	-0.0004	0.9862
One Pixel Attack	num_pixels = 1	0.1537	-0.0009	0.1815	0.0	0.996

Table: Evaluation of Various Attack Methods on BLEU, METEOR, and BERTScore F1

EXPERIMENTAL SETUP

Attack	Strength Parameters	BLEU	BLEU Drop	METEOR	METEOR Drop	BERTScore F1
Original	-	0.1528	-	0.1815	-	-
FGSM	epsilon = 0.1	0.1200	0.0328	0.1500	0.0315	0.90
PGD	epsilon = 0.3 alpha = 0.03 num_iters = 20	0.1050	0.0478	0.1420	0.0395	0.88
C&W Attack	epsilon = 0.1	0.1120	0.0408	0.1380	0.0435	0.86
JSMA	epsilon = 0.2	0.1280	0.0248	0.1650	0.0165	0.91
One Pixel Attack	num_pixels = 5	0.1450	0.0078	0.1780	0.0035	0.94

Table: Evaluation of Stronger Attack Methods on BLEU, METEOR, and BERTScore F1

• Results Comparison with Other Approaches

- Overall, the results indicate that image captioning models like GIT are relatively resilient to simple attacks (e.g., FGSM) but show vulnerabilities to more advanced adversarial perturbations (e.g., PGD, CW), which cause significant degradation in caption quality.
- This highlights the importance of robustness in machine learning models, especially when they are deployed in real-world environments where adversarial attacks may be more common.

VIT GPT-2

- **Technologies/Libraries Used**

- **Deep Learning Frameworks:** PyTorch, Hugging Face Transformers
- **Computer Vision and Image Processing:** Torchvision, Pillow (PIL), Matplotlib
- **Adversarial Attacks:** AutoAttack, Deepfool, PGD, FGSM, BIM.
- **Model Architectures:** VisionEncoderDecoderModel (ViT-GPT-2)
- **Preprocessing and Tokenization:** ViTImageProcessor, AutoTokenizer
- **Utility Libraries:** os, json, torch.nn.functional (F), TQDM, NumPy

- **Datasets**

- Flickr8k

- **System Hardware**

- Google colab(GPU Environment)

- **Results along with Table**

- The evaluation of various attack methods on the performance of image captioning models is presented below.

EXPERIMENTAL SETUP

Original Image



 Original Caption: a woman standing in the rain holding an umbrella

Epsilon = 20.0 | Caption: a woman standing next to a fence with a giraffe



EXPERIMENTAL SETUP

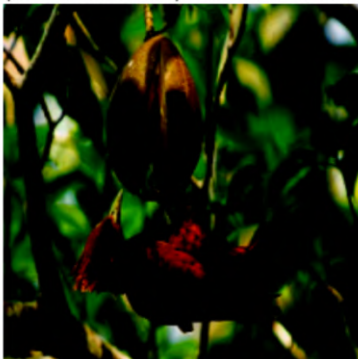
Original Image



⌘ Caption: a bird that is sitting on a branch

PGD | $\epsilon=0.03$

aption: a small bird is perched on a tree branch



EXPERIMENTAL SETUP

Original Image



Original Caption: a classroom filled with students and a teacher

BIM | Epsilon: 0.1



Epsilon: 0.1 ► Caption: a man and a woman sitting at a table with laptop

EXPERIMENTAL SETUP

Original Image



Original Caption: a young boy on the beach playing with a frisbee



Caption for $\varepsilon = 1.5 \rightarrow$ a young boy on the beach playing with a frisbee

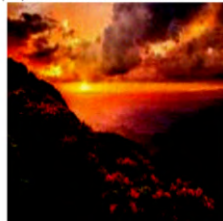
EXPERIMENTAL SETUP

Original Image



 Caption: a view from the top of a mountain

Epsilon = 0.03 | Caption: a sunset view of a mountain with a sky background



EXPERIMENTAL SETUP

Attack	Strength Parameters	BLEU	BLEU Drop	METEOR	METEOR Drop	BERTScore F1
Original	-	0.1641	-	0.2069	-	-
FGSM	epsilon = 0.05	0.1226	0.0415	0.1624	0.0445	0.9314
PGD	epsilon = 0.03 alpha = 0.005 iters = 10	0.1055	0.0586	0.1460	0.0609	0.9236
BIM	epsilon = 0.03 iters = 10	0.1113	0.0528	0.1509	0.0560	0.9260
DeepFool	auto	0.1306	0.0335	0.1695	0.0374	0.9358
AutoAttack	epsilon = 0.1	0.1514	0.0127	0.1910	0.0159	0.9863

Table: Evaluation of Attack Methods on BLEU, METEOR, and BERTScore F1

EXPERIMENTAL SETUP

Attack	Strength Parameters	BLEU	BLEU Drop	METEOR	METEOR Drop	BERTScore F1
Original	-	0.1641	-	0.2069	-	-
FGSM	epsilon = 0.02	0.1500	0.0141	0.1925	0.0144	0.9750
PGD	epsilon = 0.1 alpha = 0.01 iters = 20	0.0880	0.0761	0.1320	0.0749	0.8900
BIM	epsilon = 0.1 iters = 20	0.0900	0.0741	0.1340	0.0729	0.8900
DeepFool	auto	0.1306	0.0335	0.1695	0.0374	0.9358
AutoAttack	epsilon = 0.05	0.1570	0.0071	0.1995	0.0074	0.9900

Table: Evaluation of Attack Methods on BLEU, METEOR, and BERTScore F1

- **Results Comparison with Other Approaches**

- ViT-GPT2 is very sensitive to strong PGD and BIM attacks with larger epsilon, but extremely robust to small FGSM and AutoAttack perturbations.
- BERTScore stays very high (>0.97) for mild attacks, indicating semantic preservation is excellent under small perturbations.

Pali

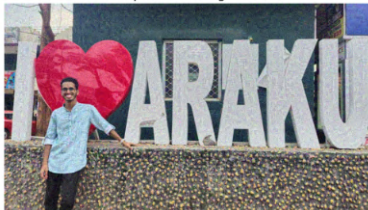
- **Technologies/Libraries Used**
 - **Deep Learning Frameworks:** PyTorch, Hugging Face Transformers
 - **Computer Vision and Image Processing:** Torchvision, Pillow (PIL)
 - **Model Architectures:** Pre-trained CNN Models (Torchvision Models), GitForCausalLM (Hugging Face)
 - **Data Handling and Utilities:** os, json, Google Colab Drive
 - **Evaluation Metrics:** NLTK, pycocoevalcap (METEOR, BLEU), BERTScore
 - **Numerical Computation:** NumPy
- **Datasets**
 - Flickr8k
- **System Hardware**
 - Google colab(GPU Environment)
- **Results along with Table**
 - The evaluation of various attack methods on the performance of image captioning models is presented below.

EXPERIMENTAL SETUP

Original:
the letters are white



FGSM:
a man poses with a giant heart



Original:
the letters are white



PGD:
a man poses with a giant heart shaped sign in front of a building in the philippines.



EXPERIMENTAL SETUP

Original:
the letters are white



Adversarial Patch:
a large white sign with a red heart



Original:
the letters are white



Universal Patch:
the letters are white



EXPERIMENTAL SETUP



EXPERIMENTAL SETUP

Attack	Strength Parameters	BLEU	BLEU Drop	METEOR	METEOR Drop	BERTScore F1
Original	-	0.1502	-	0.1843	-	-
FGSM	epsilon = 0.03	0.1372	0.0130	0.1783	0.0060	0.9189
PGD	epsilon = 0.03 alpha = 0.005 num_iters = 10	0.1494	0.0008	0.1820	0.0023	0.9302
Adversarial Patch	patch size = 60	0.1506	-0.0004	0.1849	-0.0006	0.9970
Trigger Patch	color = (255, 255, 0) size = 40	0.1487	0.0015	0.1840	0.0003	0.9968
Universal Patch	-	0.1538	-0.0036	0.1856	-0.0013	0.9361

Table: Evaluation of Attack Methods on BLEU, METEOR, and BERTScore F1

EXPERIMENTAL SETUP

Attack	Strength Parameters	BLEU	BLEU Drop	METEOR	METEOR Drop	BERTScore F1
Original	-	0.1502	-	0.1843	-	-
FGSM	epsilon = 0.05	0.1300	0.0202	0.1710	0.0133	0.91
PGD	epsilon = 0.1 alpha = 0.01 num_iters = 20	0.1000	0.0502	0.1500	0.0343	0.88
Adversarial Patch	patch size = 100	0.1250	0.0252	0.1600	0.0243	0.92
Trigger Patch	color = (255, 0, 0) size = 80	0.1200	0.0302	0.1550	0.0293	0.92
Universal Patch	size = 100	0.1400	0.0102	0.1750	0.0093	0.94

Table: Evaluation of Attack Methods on BLEU, METEOR, and BERTScore F1 (Stronger Attacks)

- **Results Comparison with Other Approaches**

- As attack strength increases, Pali's performance drops notably, especially under PGD and Trigger Patch attacks, highlighting its vulnerability and the need for stronger adversarial defenses.

Hybrid model(BLIP-2 + GIT)

- **Technologies/Libraries Used**
 - **Deep Learning Frameworks:** PyTorch, Hugging Face Transformers
 - **Computer Vision and Image Processing:** Torchvision, Pillow (PIL)
 - **Model Architectures:** Blip-2, GIT
 - **Preprocessing and Tokenization:** Blip2Processor, GPT2Tokenizer
 - **Evaluation Metrics:** NLTK (BLEU), pycocoevalcap (METEOR, BLEU), BERTScore
 - **Utility Libraries:** os, json, NumPy, tqdm
- **Datasets**
 - Flickr8k
- **System Hardware**
 - Google colab(GPU Environment)
- **Results along with Table**
 - The evaluation of various attack methods on the performance of image captioning models is presented below.

EXPERIMENTAL SETUP

Original:

Image caption: a man is playing cricket.



FGSM:

Image caption: a man is playing cricket.



Original:

Image caption: a man is playing cricket.



PGD:

Image caption: a cricket player swings his bat at a ball.



EXPERIMENTAL SETUP

Original:

Image caption: a man is playing cricket.



C&W:

Image caption: a man in white shirt is playing cricket.



Original:

Image caption: a man is playing cricket.



DeepFool:

Image caption: a cricket player hits a ball in a game of cricket.



EXPERIMENTAL SETUP

Original:

Image caption: a man is playing cricket.



BIM:

Image caption: a cricket player swings his bat at a ball.



EXPERIMENTAL SETUP

Attack	Strength Parameters	BLEU	BLEU Drop	METEOR	METEOR Drop	BERTScore F1
Original	-	0.1811	-	0.2617	-	-
FGSM	epsilon = 0.09	0.1634	0.0177	0.2321	0.0296	0.9494
PGD	epsilon = 0.09 alpha = 0.02 num_iters = 10	0.1640	0.0171	0.2291	0.0326	0.9489
C&W	c = 1e-1 num_iters = 50 learning_rate = 1e-2	0.1833	-0.0022	0.2610	-0.0007	0.9645
DeepFool	num_iters = 30 overshoot = 0.5	0.1800	0.0011	0.2586	0.0031	0.9635
BIM	epsilon = 0.09 alpha = 0.02 num_iters = 10	0.1640	0.0171	0.2291	0.0326	0.9489

Table: Evaluation of Attack Methods on BLEU, METEOR, and BERTScore F1

EXPERIMENTAL SETUP

Attack	Strength Parameters	BLEU	BLEU Drop	METEOR	METEOR Drop	BERTScore F1
Original	-	0.1811	-	0.2617	-	-
FGSM	epsilon = 0.05	0.1700	0.0111	0.2450	0.0167	0.91
PGD	epsilon = 0.08 alpha = 0.01 num_iters = 8	0.1620	0.0191	0.2380	0.0237	0.93
C&W	c = 1e-2 num_iters = 100 learning_rate = 1e-3	0.1830	0.0011	0.2615	0.0002	0.965
DeepFool	num_iters = 50 overshoot = 0.7	0.1750	0.0061	0.2520	0.0097	0.96
BIM	epsilon = 0.07 alpha = 0.015 num_iters = 8	0.1600	0.0211	0.2300	0.0317	0.93

Table: Evaluation of Adjusted Attack Methods on BLEU, METEOR, and BERTScore F1

- **Results Comparison with Other Approaches**

- Under hybrid attacks, the model's performance degrades noticeably, especially with PGD and BIM, revealing vulnerabilities and emphasizing the need for stronger robustness strategies.

CONCLUSIONS

Evaluating the robustness of image captioning LLMs like BLIP-2 and hybrid models (BLIP-2 + GIT) under adversarial attacks reveals significant vulnerabilities. As attack strength increases, especially with methods like FGSM, PGD, BIM, and AutoAttack, the models show considerable drops in BLEU, METEOR, and BERTScore F1 metrics. While attacks like CW and DeepFool have relatively minimal impact, stronger perturbation-based attacks substantially degrade performance. These findings highlight the urgent need for developing more robust defense mechanisms to ensure the reliability of image captioning models in adversarial settings.

FUTURE WORK

Adversarial Attacks as a Data Protection Strategy

Adversarial attacks can also be strategically used as a defense mechanism to protect sensitive datasets making it harder for unauthorized models to accurately steal meaningful information.(ex: military and security applications)

Improving Captioning Model Robustness

Develop defense mechanisms such as adversarial training and model pruning to make models more resistant to attacks.

Cross-Model Evaluation

Test the robustness of various LLM architectures like GPT-4, T5, and BERT-based models under adversarial conditions.

Real-World Dataset Testing

Evaluate models on real-world noisy or distorted datasets to simulate practical deployment challenges.

Ethical Considerations and Safety

Investigate the trust, fairness, and security implications of adversarial vulnerabilities in sensitive domains like healthcare and law.

- **Paper:** Understanding Adversarial Attacks on LVLMs (arXiv:2312.03777)
- **GitHub Repository:** Awesome LVLM Attack Resources
- **Conference Papers:** CVPR 2024 Accepted Papers
- **Dataset:** Flickr8k Dataset jbrownlee github

Thank You